

In our next session, we will solve these questions together in class. Here's how it will work:

- For each question, I will first ask for **volunteers** to answer.
- If no one volunteers, I will **randomly select one person** to answer.

Don't worry about making a mistake!

- If your answer is **incorrect**, no points will be deducted from your total score.
- If your answer is **correct**, you will earn **1 extra point** toward your total score.

Bonus Point Opportunity:

*You can earn up to a total of **5 extra points** during the semester. However, you cannot earn more than **1 extra point** from questions based on a single chapter.*

Chapter 3 Questions for In-Class Practice

1- Wu Systems has the following balance sheet. Assume that all current assets are used in operations. How much net operating working capital does the firm have?

Cash	\$ 100	Accounts payable	\$ 240
Accounts receivable	720	Accruals	220
Inventory	480	Notes payable	440
Current assets	\$ 1,300	Current liabilities	\$ 900
Net fixed assets	1,000	Long-term debt	650
		Common equity	250
		Retained earnings	500
Total assets	\$ 2,300	Total liab. & equity	\$ 2,300

2- Brown Fashions Inc.'s December 31, 2020 balance sheet showed total common equity of \$4,050,000 and 130,000 shares of stock outstanding. During 2021, the firm had \$390,000 of net income, and it paid out \$100,000 as dividends. What was the book value per share at 12/31/21, assuming no common stock was either issued or retired during 2021?

3- Prezas Company's balance sheet showed total current assets of \$4,250, all of which were required in operations. Its current liabilities consisted of \$1,035 of accounts payable, \$600 of 6% short-term notes payable to the bank, and \$205 of accrued wages and taxes. What was its net operating working capital?

4- Emery Mining Inc. recently reported \$130,000 of sales, \$75,500 of operating costs other than depreciation, and \$11,600 of depreciation. The company had \$16,500 of outstanding bonds that carry a 7.25% interest rate, and its federal-plus-state income tax rate was 25%. How much was the firm's net income? The firm uses the same depreciation expense for tax and stockholder reporting purposes.

5- Last year Almazan Software reported \$10.000 million of sales, \$6.750 million of operating costs other than depreciation, and \$1.200 million of depreciation. The company had \$5.000 million of bonds that carry a 6.5% interest rate, and its federal-plus-state income tax rate was 25%. This year's data are expected to remain unchanged except for one item, depreciation, which is expected to increase by \$0.650 million. By how much will net income change as a result of the change in depreciation?

6- Vasudevan Inc. recently reported operating income of \$2.70 million, depreciation of \$1.30 million, and had a tax rate of 25%. The firm's expenditures on fixed assets and net operating working capital totaled \$0.60 million. How much was its free cash flow, in millions?

7- Watson Oil recently reported (in millions) \$8,250 of sales, \$5,250 of operating costs other than depreciation, and \$1,325 of depreciation. The company had \$3,200 of outstanding bonds that carry a 5% interest rate, and its federal-plus-state income tax rate was 25%. In order to sustain its operations and thus generate future sales and cash flows, the firm was required to make \$1,250 of capital expenditures on new fixed assets and to invest \$300 in net operating working capital. By how much did the firm's net income exceed its free cash flow?

8- In 2019, Garner Grocers had taxable income of -\$7,000,000. The corporate tax rate is 25%. Assume that the company takes full advantage of the Tax Code's carry-forward provision. In 2020, Garner has taxable income of \$13,000,000. What is the amount of taxes the company paid in 2020?

9- Your corporation has the following cash flows:

Operating income	\$210,000
Interest received	\$13,000
Interest paid	\$46,000
Dividends received	\$7,000

If the applicable income tax rate is 25% (federal and state combined), and if 50% of dividends received are exempt from taxes, what is the corporation's tax liability?

10- A corporate bond currently yields 9.80%. Municipal bonds with the same risk, maturity, and liquidity currently yield 5.40%. At what tax rate would investors be indifferent between the two bonds? (Round your final answer to two decimal places.)

11- Maureen Smith is a single individual. She claims a standard deduction of \$12,400. Her salary for the year was \$115,700. Assume the following tax table is applicable.

Single Individuals

If Your Taxable Income Is	You Pay This Amount on the Base of the Bracket	Plus This Percentage on the Excess over the Base (Marginal Rate)	Average Tax Rate at Top of Bracket
Up to \$9,875	\$ 0.00	10.0%	10.0%
\$9,875–\$40,125	987.50	12.0	11.5
\$40,125–\$85,525	4,617.50	22.0	17.1
\$85,525–\$163,300	14,605.50	24.0	20.4
\$163,300–\$207,350	33,271.50	32.0	22.8
\$207,350–\$518,400	47,367.50	35.0	30.1
Over \$518,400	156,235.00	37.0	37.0

Married Couples Filing Joint Returns

If Your Taxable Income Is	You Pay This Amount on the Base of the Bracket	Plus This Percentage on the Excess over the Base (Marginal Rate)	Average Tax Rate at Top of Bracket
Up to \$19,750	\$ 0.00	10.0%	10.0%
\$19,750–\$80,250	1,975.00	12.0	11.5
\$80,250–\$171,050	9,235.00	22.0	17.1
\$171,050–\$326,600	29,211.00	24.0	20.4
\$326,600–\$414,700	66,543.00	32.0	22.8
\$414,700–\$622,050	94,735.00	35.0	26.9
Over \$622,050	167,307.50	37.0	37.0

What is her average tax rate?

12- John has the following types of income for the year 2020:

- 1) Earned income (wages, salaries, and bonuses): \$96,000
- 2) Interest income: \$9,000
- 3) Qualified dividends: \$12,000
- 4) Ordinary dividends: \$6,000
- 5) Short-term capital gains (held for less than 1 year): \$11,000
- 6) Long-term capital gains (held for more than 1 year): \$18,000
- 7) Business income from a sole proprietorship: \$20,000
- 8) John is eligible for the standard deduction of \$12,400 (for a single filer).

Using the information provided in the tax table from Question 11, calculate John's total tax liability for the year 2020.